

☹ You CANNOT use this or any other formula sheet on the Assessment. You need to memorize this stuff

## Algorithms

Here's a list of some functions listed from fast to slow:

$\mathcal{O}(1)$  **Constant**

$\mathcal{O}(\log(n))$  **Logarithmic.**

Ignore Log bases:  $\log_{10}(n) = \log(n)$

Same growth as  $\log(n^k) = k \log(n)$

$\mathcal{O}(\log^c(2n))$  **Polylogarithmic**

Don't forget that:  $\log^c x = (\log x)^c$ .

$\mathcal{O}(n)$  **Linear**

$\mathcal{O}(n \log(2n))$  **Linearithmic/Loglinear**

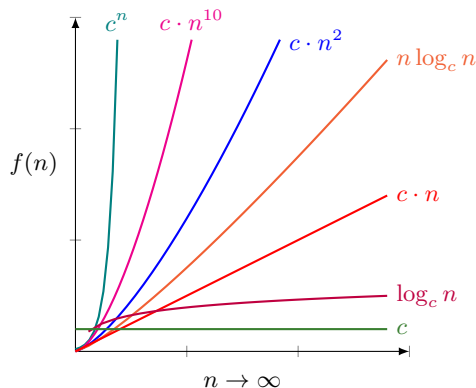
$\mathcal{O}(n^2)$  **Quadratic**

$\mathcal{O}(n^2 \log(2n))$

$\mathcal{O}(n^3)$  **Polynomial**

$\mathcal{O}(2^n)$  **Exponential**

$\mathcal{O}(n!)$



### Some properties

We ignore coefficients and log bases, but not exponent bases.

ex.  $3 \cdot 5^x$  is  $\mathcal{O}(5^x)$

When adding functions, we only take the slowest function's Big- $\mathcal{O}$ .

ex.  $0.01x^3 + 12x^2 - 2 \log 37x$  is  $\mathcal{O}(n^3)$

When multiplying functions, we take the product of the functions' Big- $\mathcal{O}$ .

ex.  $x^2 + 2 \in \mathcal{O}(n^2)$  and  $\log_{10}(x) \in \mathcal{O}(\log n)$  so then  $(x^2 + 2)(\log_{10}(x))$  is  $\mathcal{O}(n^2 \log(n))$

## Advanced Counting

**Combination** Number ways to choose  $k$  objects from a group of  $n$  when *order doesn't matter*:

$${}_nC_k = \binom{n}{k} = \frac{n!}{n-k!k!}$$

**Permutation** Number ways to choose  $k$  objects from a group of  $n$  when *order does matter*:

$${}_nP_k = \frac{n!}{n-k!}$$

### Binomial Theorem

$$(a+b)^n = \sum_{k=0}^n \binom{n}{k} a^k b^{n-k}$$

This gives us the following identity:

$$\binom{n}{0} - \binom{n}{1} + \binom{n}{2} - \cdots (-1)^n \binom{n}{n} = 0$$

## Discrete Probability

- $P(A)$  is the probability of event  $A$  happening.
- $P(\bar{A}) = 1 - P(A)$  is the probability of event  $A$  **not** happening.
- $P(A|B)$  is the probability of event  $A$  *given* that  $B$  has already happened. When  $A$  and  $B$  are independent,  $P(A|B) = P(A)$ .
- $P(A \cap B) = P(A|B)P(B)$  is the probability of event  $A$  *and*  $B$  happening.
- $P(A \cup B) = P(A) + P(B) - P(A \cap B)$  is the probability of the  $A$  *or*  $B$  happening.

**Bayes' Theorem:** For events  $X$  and  $F$  from a common sample space where  $P(X), P(F) \neq 0$ ,

$$P(F|X) = \frac{P(X|F)P(F)}{P(X)} \quad (\text{version 1})$$

and equivalently,

$$P(F|X) = \frac{P(X|F)P(F)}{P(X|F)P(F) + P(X|\bar{F})P(\bar{F})} \quad (\text{version 2})$$

While the first version does not appear in our textbook or pre-assessment practice problems, this equivalent definition is listed in the lesson summary.

Note that  $P(F \cap X) = P(X|F)P(F)$ . So version 1 is often shown as:

$$P(F|X) = \frac{P(X \cap F)}{P(X)} \quad (\text{version 3})$$

**Expected Value** The *expected value* of a discrete random variable  $X$  is

$$E[X] = \sum_{s \in S} X(s)p(s)$$

where  $S$  is the set of possible outcomes of  $X$ ,  $p(s)$  is the probability of outcome  $s$ , and  $X(s)$  is the value of  $X$  for  $s$ .

## Number Theory

Stuff to add?

- modular arithmetic basics
- needed modular arithmetic properties
- Euclidean algorithm
- Extended Euclidean Algorithm
- Encryption

## Recursion and Induction

Stuff to add?

- Label for Induction parts
- Summation formulation label?